

Math 237A Final

Zih-Yu Hsieh

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Problem 1

Lazarsfeld Problem Set 6 (4):

Let $\mathbb{G} = \mathbb{G}(1, 3)$ be the Grassmannian parametrizing lines in $\mathbb{P}^3$, and fix a line $\ell_0 \subset \mathbb{P}^3$. Let $\Sigma \subset \mathbb{G}$ be the subset of $\mathbb{G}$ corresponding to all lines meeting ℓ_0 , so that (as we've seen before) Σ is a hyperplane section of $\mathbb{G}$.

- Show that Σ has a unique singular point (at the point of $\mathbb{G}$ corresponding to ℓ_0). In fact, Σ is a quadric Q of rank 4 in $\mathbb{P}^4$.
- Let $Q \subset \mathbb{P}^4$ be a quadric of rank 4 in $\mathbb{P}^4$, with singular point $o \in Q$. Let Q' be the proper transform of Q under the blowing up of $o \in \mathbb{P}^4$. Show that Q' is non-singular, and that the fibre of $\nu : Q' \rightarrow Q$ over o is (isomorphic to) $\mathbb{P}^1 \times \mathbb{P}^1$.
- Returning to the realization Σ of Q in (a), define $\Sigma' \subseteq \mathbb{G} \times \ell_0$ to be the incidence correspondance $\Sigma' = \{([\ell], x) \mid x \in \ell \cap \ell_0\}$. Show that Σ' is smooth (non-singular), and that the projection $\mu : \Sigma' \rightarrow \Sigma$ is an isomorphism away from $[\ell_0]$, while $\mu^{-1}[\ell_0]$ is a copy of $\mathbb{P}^1$. This map is called the „small resolution“ of $Q = \Sigma$.

Solution:

- First, recall that the Grassmannian has Plucker Embedding $\mathbb{G}(1, 3) \hookrightarrow \mathbb{P}^5$, where for each line (or a 2-dimensional subspace in k^4), a corresponding matrix $\begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{pmatrix}$ in $\mathbb{P}^5$ corresponds to the point $[X_{12}, X_{13}, X_{14}, X_{23}, X_{24}, X_{34}] \in \mathbb{P}^5$, where each $X_{ij} = \det \begin{pmatrix} x_i & x_j \\ y_i & y_j \end{pmatrix}$.

WLOG, up to a projective linear transform one can assume the point $\ell_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$, which corresponds to the point $[1, 0, 0, 0, 0, 0] \in \mathbb{G}(1, 3) \subset \mathbb{P}^5$. Which, we claim that a line $\ell = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{pmatrix} \in \Sigma$ iff $X_{34} = \det \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = 0$.

$\Rightarrow$: If $\ell \in \Sigma$, then it intersects with ℓ_0 . On the level of k^4 , it implies there exists $[s, t] \in \mathbb{P}^1$, such that $s[x_1, x_2, x_3, x_4] + t[y_1, y_2, y_3, y_4] \in \ell_0$, which has the last two coordinates being zero, showing $(sx_3 + ty_3, sx_4 + ty_4) = (s \ t) \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = (0 \ 0)$. With $(s, t) \neq 0$ as a vector, this implies $\det \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = 0$, hence the corresponding Pucker coordinate has $X_{34} = 0$.

$\Leftarrow$: If $X_{34} = \det \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = 0$, there exists nonzero (s, t) , such that $(s \ t) \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = (sx_3 + ty_3, sx_4 + ty_4) = 0$. Hence, the vector (in k^4) / point (in $\mathbb{P}^3$) has $[sx_1 + ty_1, sx_2 + ty_2, 0, 0] \in \ell_0$, showing $\ell \cap \ell_0 \neq \emptyset$, or $\ell \in \Sigma$.

Hence, $\Sigma \subset \mathbb{G}(1, 3) \subset \mathbb{P}^5$ is the intersection $\{X_{34} = 0\} \cap \mathbb{G}(1, 3)$; with $\mathbb{G}(1, 3)$ being cut out by the equation $X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23} = 0$, the defining equation for $\Sigma \subset \mathbb{P}^5$ is $\{X_{34} = 0, X_{14}X_{23} - X_{13}X_{24} = 0\}$. And, notice that since the last coordinate is 0, one can simply project Σ down to $\mathbb{P}^4$ with coordinates $[X_{12}, X_{13}, X_{14}, X_{23}, X_{24}]$, and now the defining equation is $X_{14}X_{23} - X_{13}X_{24} = 0$.

For this equations, the partial derivatives all vanished iff $X_{14} = X_{23} = X_{13} = X_{24} = 0$ (since the partials would precisely result in these for, up to ± 1), showing the only singular

point must be at $[X_{12}, 0, 0, 0, 0] = [1, 0, 0, 0, 0]$. Which, the only line in $\mathbb{P}^3$ with $X_{12} \neq 0$, is ℓ_0 . So, ℓ_0 is the only singular point.

- (b) WLOG, up to change of coordinates, say the singularity occurs at $o = [1, 0, 0, 0, 0]$ (and say $\mathbb{P}^4$ has coordinates $[x_0, x_1, x_2, x_3, x_4]$). Then, if say Q is a quadric defined by homogeneous equation f , then within the affine chart U_{x_0} (where $x_0 \neq 0$, WLOG say $x_0 = 1$), the equation of Q is defined by $f(1, x_1, x_2, x_3, x_4)$, and must have all of its partial derivatives vanished at $x_1 = x_2 = x_3 = x_4 = 0$. This indicates that one can't have any terms with x_0 involved. Which, f can be viewed as a homogeneous polynomial in $x_1, \dots, x_4$.

As a result, when blowing up with respect to this affine chart (say $U_{x_0} \cong \mathbb{A}^4$ with coordinates (x_1, x_2, x_3, x_4)), then one has $\text{Bl}_0 U_{x_0} = \left\{ \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ W & X & Y & Z \end{pmatrix} \right\}$ (where $[W, X, Y, Z] \in \mathbb{P}^3$ and the matrix has rank ≤ 1). Which, if pick the affine chart U_W (where $W \neq 0$), then within the blowing-up the proper transform of Q , say Q' , has the equation of Q pulling back to $f(1, x_1, x_1 X, x_1 Y, x_1 Z) = x_1^2 \cdot f(1, 1, X, Y, Z)$ (since f is homogeneous, and for quadric surface the degree is 2), where x_1^2 cuts out the exceptional divisor in this chart, so the defining equation of Q' in U_W is $f(1, 1, X, Y, Z)$. Which, notice that since within $Q \subset \mathbb{P}^4$, the only singularity is o , then with the point $[1, 1, X, Y, Z] \neq o$, hence any partial derivatives with respect to X, Y, Z has f being non-vanishing, showing it's non-singular. If apply similar results, this indicates that it is non-singular at all points.

(Isomorphism to $\mathbb{P}^1 \times \mathbb{P}^1$ now is unclear).

- (c) For $\ell \in \Sigma$ such that $\ell \neq \ell_0$, it's clear that $\ell \cap \ell_0 = \{x\}$ a single point, so there exists a unique $x \in \ell_0$, where $([\ell], x) \in \Sigma'$. Hence, restricting to $\Sigma \setminus \{[\ell_0]\}$, the map $\psi : \Sigma \setminus \{[\ell_0]\} \rightarrow \Sigma'$ has $\psi([\ell]) = ([\ell], \ell \cap \ell_0)$ being a well-defined morphism (since it requires the second coordinate to satisfy the equation for both ℓ and ℓ_0 as a variety in $\mathbb{P}^3$), and it defines an isomorphism onto its image (since each $\ell \neq \ell_0$ corresponds to a unique intersection). Which, the inverse is instead given by the projectio (since $\mu \circ \psi([\ell]) = \mu([\ell], \ell \cap \ell_0) = [\ell]$, while $([\ell], x)$ with $x = \ell \cap \ell_0$ has $\psi \circ \mu([\ell], x) = \psi([\ell]) = ([\ell], x)$).

Finally, the reason why $\mu^{-1}([\ell_0]) = \mathbb{P}^1$, because for any $x \in \ell_0$, one has $x \in \ell_0 \cap \ell_0$, hence $([\ell_0], x) \in \Sigma'$. Which, one has $\mu^{-1}([\ell_0]) = \{([\ell_0], x) \mid x \in \ell_0\}$ has a one-to-one correspondence to ℓ_0 by projecting down to the second coordinates. Then, since $\ell_0 \cong \mathbb{P}^1$ (a copy of $\mathbb{P}^1$), then $\mu^{-1}([\ell_0]) \cong \mathbb{P}^1$.

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Problem 2

Lazarsfeld Problem Set 7 (1):

Let $X \subseteq \mathbb{C}^3$ be the hypersurface $x^2 + y^4 + z^4 = 0$, and denote $X' \subseteq \text{Bl}_0(\mathbb{C}^3)$ the proper transform of X under the blowing up of the origin. Show that X has an isolated singularity, but that X' is singular along a curve.

Solution:

I. Isolated Singularity of X :

Consider the partial derivatives of the defining equation $f(x, y, z) = x^2 + y^4 + z^4$, one has $\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = 4y^3$, and $\frac{\partial f}{\partial z} = 4z^3$. Which, the singularity happens when all three partial derivatives vanished, which requires $2x = 4y^3 = 4z^3 = 0$, showing $x = y = z = 0$. Hence, the only singularity of X is $(x, y, z) = (0, 0, 0)$.

II. Singularity of Proper Transform:

If consider the blowing up of $\mathbb{C}^3$ at the origin, it is given by $\text{Bl}_0 \mathbb{C}^3 \subset \mathbb{C}^3 \times \mathbb{P}^2$, where $\text{Bl}_0 \mathbb{C}^3 = \{(\bar{x}, [\ell]) \mid \bar{x} \in \ell\}$ (where $\bar{x} = (x, y, z)$ is a point of the 1-dimensional subspace $\ell \subset \mathbb{C}^3$). If expresses $[\ell] = [X', Y, Z]$, then $\bar{x} \in \ell$ implies (x, y, z) is some multiple of (X', Y, Z) , hence the matrix $\begin{pmatrix} x & y & z \\ X' & Y & Z \end{pmatrix}$ has rank ≤ 1 . Which, on the open set $Z \neq 0$ (WLOG $Z = 1$), $\text{Bl}_0 \mathbb{C}^3$ has $(\bar{x}, [\ell])$ satisfy $\bar{x} = (x, y, z)$ and $[\ell] = (X, Y, 1)$ being linearly dependent, showing $x = zX$ and $y = zY$. Which, it can be described by the affine chart (X, Y, z) , with projection onto $\mathbb{C}^3$ given by $(X, Y, z) \mapsto (Xz, Yz, z)$.

Notice that under this affine chart, the defining equation $f = x^2 + y^4 + z^4$ pulls back to $z^2 X^2 + z^4 Y^4 + z^4 = z^2(X^2 + z^2 Y^4 + z^2) = 0$. Which, excluding the exceptional divisor's equation z^2 , the defining equation is $f_z = X^2 + z^2 Y^2 + z^2 = 0$. Which, to find singular points, its partial derivatives are given as follow:

$$\frac{\partial f_z}{\partial X} = 2X, \quad \frac{\partial f_z}{\partial Y} = 2YZ^2, \quad \frac{\partial f_z}{\partial z} = 2z(1 + Y^2) \quad (2.1)$$

Which, if all three partial derivatives vanish, one has $X = 0$ (for ∂X), $Y = 0$ or $z = 0$ (for ∂Y), $z = 0$ or $Y^2 = -1$ (for ∂z).

Which, all the points of the form $X = z = 0$ has all partial derivatives vanished, also $f_z(0, Y, 0) = 0^2 + 0^2 \cdot Y^2 + 0^2 = 0$, showing these points are contained in the proper transform. Hence, the proper transform is non-singular, and within the affine chart (where $Z \neq 0$), one has all $(0, Y, 0)$ (where $Y \in \mathbb{C}$) being singular. Hence, the proper transform is singular along a curve.

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Problem 3

Hartshorne 5.1:

Locate the singular points and sketch the following curves in $\mathbb{A}^2$ (assume $\text{char}(k) \neq 2$). Which is which in Figure 4?

- (a) $x^2 = x^4 + y^4$
- (b) $xy = x^6 + y^6$
- (c) $x^3 = y^2 + x^4 + y^4$
- (d) $x^2y + xy^2 = x^4 + y^4$

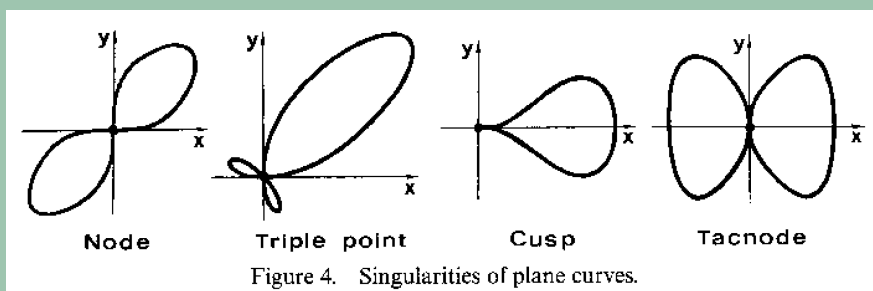


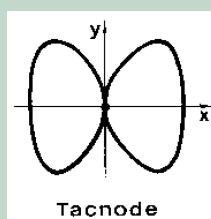
Figure 4. Singularities of plane curves.

Solution:

- (a) Given the equation $f(x, y) = x^4 - x^2 + y^4$, the singular points occurred at where all the partial derivatives are 0. Which, one has $\frac{\partial f}{\partial x} = 4x^3 - 2x = 0$ (which occurred at $x = 0$, and $4x^2 - 2 = 0 \Rightarrow x^2 = \frac{1}{2}$); also, one has $\frac{\partial f}{\partial y} = 4y^3 = 0$ (which occurred at $y = 0$). Hence, the points where all partial derivatives vanished include $(x, y) = (0, 0), (z_1, 0), (z_2, 0)$ (where $z_1, z_2 \in k$ satisfy $z_1^2 = z_2^2 = \frac{1}{2}$).

Out of these points, only $(0, 0)$ is a solution to $f(x, y)$ (since if $z = z_1, z_2$, one has $z^2 = \frac{1}{2}$, so $f(z, 0) = z^4 - z^2 + 0^4 = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} \neq 0$). So, the only singular point on the curve is $(0, 0)$.

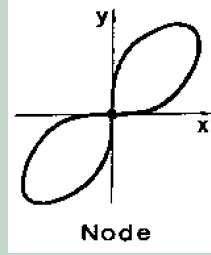
Which, if limited to $k = \mathbb{R}$, given that (x, y) is a solution to f , one also has $(x, -y), (-x, y), (-x, -y)$ being solutions to f (since all x, y in f has even power, so the -1 doesn't impact the solution), this implies that x is invariant under reflection over x and y axis, showing the only possible graph is the Tacnode.



- (b) Given the equation $g(x, y) = x^6 - xy + y^6$, the singular points occurred at where all partial derivatives are 0. Which, $\frac{\partial g}{\partial x} = 6x^5 - y = 0$ (where $y = 6x^5$), and $\frac{\partial g}{\partial y} = 6y^5 - x = 0$ (where $x = 6y^5$). This indicates that $x = 6^6 \cdot x^{25}$, so either $x = 0$ (indicating $y = 0$ also), or $x^{24} = \frac{1}{6^6}$ (which this has no inconsistency; notice that y also satisfies $6^6 \cdot y^{25} = y$, so $y = 0$ or $y^{24} = \frac{1}{6^6}$; with $y = 6x^5$, $y^{24} = 6^{24} \cdot (x^{24})^5 = 6^{24} \cdot (\frac{1}{6^6})^5 = \frac{6^{24}}{6^{30}} = \frac{1}{6^6}$). So, points where partial derivatives all vanished include $(x, y) = (0, 0)$, and $(x, y) = (z, 6z^5)$ (where z is a solution to the equation $z^{24} = \frac{1}{6^6}$).

Out of these points, the only one that is a solution to $g(x, y)$ is $(0, 0)$ (since $(z, 6z^5)$ has $f(z, 6z^5) = z^6 - z \cdot 6z^5 + (6z^5)^6 = z^6 - 6z^6 + 6^6 \cdot z^{24} \cdot z^6 = z^6 - 6z^6 + z^6 = -4z^6 \neq 0$). So, the only singular point on the curve is $(0, 0)$.

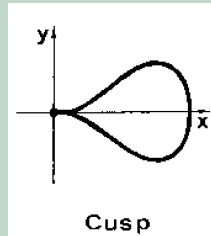
If limited to $k = \mathbb{R}$ again, if (x, y) is a solution to g , one also has $(-x, -y)$ being a solution to g (since the homogeneous degrees of each monomials are all even). Hence, the only possible curves are the Node or the Tacnode (since it's invariant under rotation by radian π); yet, because $(-x, y)$ is not a solution for most $x, y \neq 0$ (since if (x, y) is a solution to g , one has $x^6 + y^6 = xy \neq 0$, then $x^6 + y^6 \neq -xy$, showing $(-x)^6 - (-x)y + y^6 \neq 0$). So, the curve is not invariant under the reflection over y -axis, showing it's not a Tacnode. Hence, it must be a Node.



- (c) Given the equation $h(x, y) = x^4 - x^3 + y^4 + y^2$, the singular points occurred at where all partial derivatives vanishes. Which, $\frac{\partial h}{\partial x} = 4x^3 - 3x^2 = 0$ (where $x = 0$ or $x = \frac{3}{4}$), and $\frac{\partial h}{\partial y} = 4y^3 + 2y = 0$ (where $y = 0$, or $y^2 = -\frac{1}{2}$). Hence, let $z_1, z_2 \in k$ satisfy $z_1^2 = z_2^2 = -\frac{1}{2}$, the points where all partial derivatives vanished include $(0, 0)$, $(0, z_1)$, $(0, z_2)$, $(\frac{3}{4}, 0)$, $(\frac{3}{4}, z_1)$, and $(\frac{3}{4}, z_2)$.

However, only $(0, 0)$ is actually a solution to $h(x, y)$ (for $(0, z_1)$, $(0, z_2)$, and $(\frac{3}{4}, 0)$, it's clear that plugging in creates equation of only x or y , but none of the numbers are solutions; also, $h(\frac{3}{4}, z_1) = \frac{81}{256} - \frac{27}{64} + \frac{1}{4} - \frac{1}{2} \neq 0$). So, the only singular point on the curve is $(0, 0)$.

If limit to $k = \mathbb{R}$, if (x, y) is a solution to h , then $(x, -y)$ is also a solution to h (since all terms include y with even degree), this indicates the solution curve is invariant under reflection by x -axis, which can only be a Cusp or a Tacnode; yet, in general for $x, y \neq 0$, if (x, y) is a solution, $(-x, y)$ is not (for instance, take $(x, y) = 1, 0$, one has $h(1, 0) = 1^4 - 1^3 + 0^4 + 0^2 = 0$, yet $h(-1, 0) = (-1)^4 - (-1)^3 + 0^4 + 0^2 = 2 \neq 0$), so the curve is not invariant under reflection by y -axis, showing it can't be a Tacnode. Hence, it's a Cusp.



- (d) For $l(x, y) = x^4 - x^2y - xy^2 + y^4$, the singular points occurred at where all partial derivatives vanished. Which, $\frac{\partial l}{\partial x} = 4x^3 - 2xy - y^2 = 0$, and $\frac{\partial l}{\partial y} = 4y^3 - x^2 - 2xy = 0$. Which, using these equations, a singular on a curve also needs to satisfy $y^2 = 4x^3 - 2xy$, and $y^3 = \frac{x^2}{4} + \frac{xy}{2}$. Plugin $y^2 = 4x^3 - 2xy$ to $l(x, y)$, one has the singular points satisfy:

$$-3x^4 + x^2y + y^4 = 0 \quad (3.1)$$

Now, substituting $y^3 = \frac{x^2}{4} + \frac{xy}{2}$ in, one also has:

$$-4x^4 + x^2y = 0 \implies x^2(y - 4x^2) = 0 \quad (3.2)$$

So, there are two outcomes:

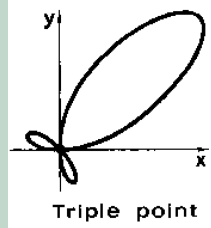
- If $x = 0$, this enforces $y = 0$ (since $y^2 = 4x^3 - 2xy = 0$). Which, $(0, 0)$ is indeed a singular point (since $l(0, 0) = 0$, while the partial derivatives also vanished).
- Else, if $x \neq 0$, then $y = 4x^2$ instead. Plugin this to $\frac{\partial l}{\partial x}$, one has:

$$4x^3 - 8x^3 - 16x^4 = 0 \implies -4x^3(4x + 1) = 0 \quad (3.3)$$

Hence, either $x = 0$ (which is ruled out in this case), or $x = -\frac{1}{4}$. Then, $y = 4x^2$ implies $y = \frac{1}{4}$. Yet, $(x, y) = (-\frac{1}{4}, \frac{1}{4})$ is not a singular point, since $l(-\frac{1}{4}, \frac{1}{4}) = \frac{1}{256} - \frac{1}{64} + \frac{1}{64} + \frac{1}{256} = \frac{1}{128} \neq 0$, so it's not on the curve.

Hence, the only singular point on the curve is $(0, 0)$.

When limit to $k = \mathbb{R}$, notice that if (x, y) is a solution to l , then so is (y, x) (since each $x^p y^q$ involved in $l(x, y)$, one also has $y^p x^q$ involved in it), so it's invariant under reflection about the line $y = x$, so the only possibilities are Node and Triple Point. Yet, in general if (x, y) is a solution, $(-x, -y)$ need not be a solution: Take $(x, y) = (1, 1)$, $l(1, 1) = 1 - 1 - 1 + 1 = 0$, yet $l(-1, -1) = 1 + 1 + 1 + 1 = 4 \neq 0$, showing $(-1, -1)$ is not a solution. Hence, it's not invariant under the rotation by radian π , which can't be a Node. So, it must be a Triple Point.



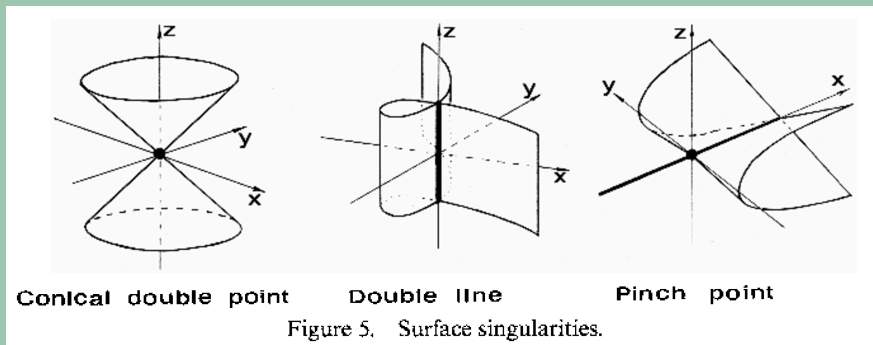
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Problem 4

Hartshorne 5.2:

Locate the singular points and describe the singularities of the following surfaces in $\mathbb{A}^3$ (assume $\text{char}(k) \neq 2$). Which is which in Figure 5?

- (a) $xy^2 = z^2$
- (b) $x^2 + y^2 = z^2$
- (c) $xy + x^3 + y^3 = 0$

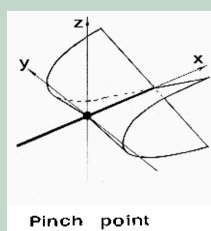


Solution:

- (a) Given $f(x, y, z) = xy^2 - z^2$, its singular points occurred as all partial derivatives vanished. Which, $\frac{\partial f}{\partial x} = y^2 = 0$ implies $y = 0$, $\frac{\partial f}{\partial y} = 2xy = 0$ implies $x = 0$ or $y = 0$, and $\frac{\partial f}{\partial z} = -2z = 0$ implies $z = 0$. So, the partial derivatives all vanished when $(x, y, z) = (x, 0, 0)$.

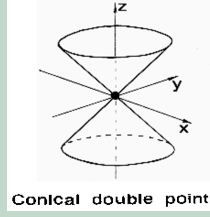
Also, for any points of this form, it satisfies $f(x, 0, 0) = x \cdot 0^2 - 0^2 = 0$, hence any point of this form is on the surface, which is a singular point. So, all singular points are of the form $(x, 0, 0)$ (where $x \in k$).

If consider $k = \mathbb{R}$, then since the surface has a whole line worth of singularities, it must either be a double line, or a Pinch point; also, since if (x, y, z) is a solution, one also has $(x, -y, z)$ being a solution (since the term involving y has even power of y). Hence, the curve is invariant when reflecting by the xz -plane, which cannot be the Double line in this case. Hence, it is the Pinch point.



- (b) Given $g(x, y, z) = x^2 + y^2 - z^2$, its singular points occurred where all partial derivatives vanished. So, $\frac{\partial g}{\partial x} = 2x = 0$ (implying $x = 0$), $\frac{\partial g}{\partial y} = 2y = 0$ (implying $y = 0$), and $\frac{\partial g}{\partial z} = 2z = 0$ (implying $z = 0$). Hence, the only possible singular point is $(0, 0, 0)$ (which is also on the curve, so it's actually a singular point).

for $k = \mathbb{R}$, since it only has a singular point, it must be the Conical Double Point (also, for each fixed z , x, y satisfy the equation for a circle, hence the intersection with a horizontal plane is always a circle).



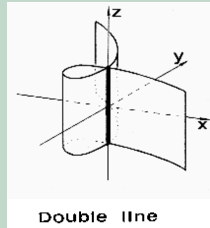
- (c) Given $h(x, y, z) = xy + x^3 + y^3$, the singular points occurred when the partial derivatives are all vanishing. Hence, $\frac{\partial h}{\partial x} = y + 3x^2 = 0$, showing $y = -3x^2$; $\frac{\partial h}{\partial y} = x + 3y^2 = 0$, showing $x = -3y^2$; $\frac{\partial h}{\partial z} = 0$, which doesn't impose extra restriction. Plug in the equations to $h(x, y, z)$, one gets the following:

$$-3y^3 - 27y^6 + y^3 = 0 \implies y^3(27y^3 + 2) = 0 \quad (4.1)$$

S, either $y = 0$ (which implies $x = 0$), or $y^3 = -\frac{2}{27}$. Which, the two potential collections of singular points, are $(0, 0, z)$ (where $z = k$), or $(-3y^2, y, z)$ (where $z \in k$, and y satisfies $y^3 = -\frac{2}{27}$).

The first collection $(0, 0, z)$ is guaranteed to be singular points, since $h(0, 0, z) = 0$ (all entries are x, y), while the second one is not singular (since if $y^3 = -\frac{2}{27}$ and $x = -3y^2$, with the equation $y = -3x^2$, one needs $y = -27y^4$, so $y(27y^3 + 1) = 0$, hence $y \neq 0$ implies $y^3 = -\frac{1}{27}$, which contradicts the assumption that $y^3 = -\frac{2}{27}$). So, the only singular points are $(0, 0, z)$ (where $z \in k$).

Consider $k = \mathbb{R}$, then since the whole z -axis is singular, then it must be the Double line.



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Problem 5

Hartshorne 5.3:

Multiplicities. Let $Y \subseteq \mathbb{A}^2$ be a curve defined by the equation $f(x, y) = 0$. Let $P = (a, b)$ be a point of $\mathbb{A}^2$. Make a linear change of coordinates so that P becomes the point $(0, 0)$. Then write f as a sum $f = f_0 + f_1 + \dots + f_d$, where f_i is a homogeneous polynomial of degree i in x and y . Then we define the *multiplicity* of P on Y , denoted $\mu_P(Y)$, to be the least r such that $f_r \neq 0$ (Note that $P \in Y \iff \mu_P(Y) > 0$). The linear factors of f_r are called the *tangent directions* at P .

- (a) Show that $\mu_P(Y) = 1 \iff P$ is a nonsingular point of Y .
- (b) Find the multiplicity of each of the singular points in Exercise 5.1 (**Problem 3**) above.

Solution:

- (a) First, one knows that P is nonsingular $\iff$ at least one of the partial derivatives of f is non-vanishing (so, one of $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ is nonzero at $P = (0, 0)$). Which, this is equivalent to having one of the $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ endowed with nonzero constant term.

Which, notice that given any homogeneous monomial $x^p y^q$ of degree $p + q$, its partial derivative of x is $p \cdot x^{p-1} y^q$ with degree $(p + q - 1)$ (if the result is nonzero), while its partial derivative of y is $q \cdot x^p y^{q-1}$ with degree $(p + q - 1)$ again (if the result is nonzero). Hence, given a homogeneous polynomial f_n of degree n , if its partial derivative (with respect to any variable x_i) is nonzero, then $\frac{\partial f_n}{\partial x_i}$ is of homogeneous degree $n - 1$. Hence, $\frac{\partial f}{\partial x} = \sum_{i=1}^d \frac{\partial f_i}{\partial x}$ and $\frac{\partial f}{\partial y} = \sum_{i=1}^d \frac{\partial f_i}{\partial y}$, where each nonzero $\frac{\partial f_i}{\partial x}, \frac{\partial f_i}{\partial y}$ is of homogeneous degree $i - 1$.

$\implies$: If $\mu_P(Y) = 1$, this implies the term $f_1 \neq 0$, hence $f_1 = ax + by$ for some $a, b \in k$ (where one of them is nonzero). For definiteness, say $a \neq 0$, hence $\frac{\partial f}{\partial x}$ contains constant term $\frac{\partial f_1}{\partial x} = a \neq 0$, showing $\frac{\partial f}{\partial x} \neq 0$ at $P = (0, 0)$.

$\impliedby$: Suppose P is a nonsingular point, then we know one of the $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ has nonzero constant term. For definiteness say $\frac{\partial f}{\partial x} = \sum_{i=1}^d \frac{\partial f_i}{\partial x}$ has nonzero constant term, then $\frac{\partial f_1}{\partial x} = a$ for some nonzero $a \in k$. Hence, one must have $f_1 = ax + by$ for some $b \in k$, showing $f_1 \neq 0$, or $\mu_P(Y) = 1$ (since $P \in Y$ implies $\mu_P(Y) > 0$ implies $f_0 = 0$, so $f_1 \neq 0$ implies $\mu_P(Y) = 1$).

- (b) In 5.1 (**Problem 3**), it turns out that all four curves in $\mathbb{A}^2$ only contain $(0, 0)$ as their only singular point. Which, one has the following:
 - For Y defined by $x^4 + y^4 - x^2 = 0$, the smallest degree homogeneous polynomial is $-x^2$, hence $\mu_{(0,0)}(Y) = 2$.
 - For Y defined by $x^6 + y^6 - xy = 0$, the smallest degree homogeneous polynomial is $-xy$, hence $\mu_{(0,0)}(Y) = 2$.
 - For Y defined by $x^4 + y^4 - x^3 + y^2 = 0$, the smallest degree homogeneous polynomial is $-y^2$, hence $\mu_{(0,0)}(Y) = 2$.
 - For Y defined by $x^4 + y^4 - x^2 y - xy^2 = 0$, the smallest degree homogeneous polynomial is $-x^2 y - xy^2$, hence $\mu_{(0,0)}(Y) = 3$.

6 D

Problem 6

Hartshorne 5.6:

Blowing Up Curve Singularities:

- (a) Let Y be a cusp or node of Exercise 5.1 (**Problem 3**). Show that the curve $\tilde{Y}$ obtained by blowing up Y at $O = (0, 0)$ is non-singular (cf. Hartshorne (4.9.1) and (Exercise 4.10)).
- (b) We define a *node* (also called *ordinary double point*) to be a double point (i.e. a point of multiplicity 2) of a plane curve with distinct tangent directions (Exercise 5.3, **Problem 5**). If P is a node on a plane curve Y , show that $\varphi^{-1}(P)$ consists of two distinct nonsingular points on the blown-up curve $\tilde{Y}$. We say that „blowing up P resolves the singularity at P “.
- (c) Let $P \in Y$ be the tacnode of (Exercise 5.1, **Problem 3**). If $\varphi : \tilde{Y} \rightarrow Y$ is the blowing-up at P , show that $\varphi^{-1}(P)$ is a node. Using (b) we see that the tacnode can be resolved by two successive blowings-up.
- (d) Let Y be the plane curve $y^3 = x^5$, which has a „higher order cusp“ at O . Show that O is a triple point; that blowing up O give rise to a double point (what kind?) and that one further blowing up resolves the singularity.

Note: We will see late (V, 3.8) that any singular point of a plane curve can be resolved by a finite sequence of successive blowings-up.

Solution: We'll consider the blowing-up as the proper transform of the algebraic sets. under the blowing-up $\text{Bl}_0 \mathbb{A}^2 = \{(\bar{x}, [\ell]) \in \mathbb{A}^2 \times \mathbb{P}^1 \mid \bar{x} \in [\ell]\}$ (where $[\ell] = [X, Y]$ represents a 1-dimensional subspace in $\mathbb{A}^2$, and with $\bar{x} = (x, y)$, they satisfy $\begin{pmatrix} x & y \\ X & Y \end{pmatrix}$ has rank ≤ 1), and the projection $\varphi : \text{Bl}_0 \mathbb{A}^2 \rightarrow \mathbb{A}^2$.

- (a) **Cusp:** It's corresponding to the equation $f = x^4 - x^3 + y^4 + y^2 = 0$.

Under the open subset U_X (where $X \neq 0$, WLOG say $X = 1$), since (x, y) is a scalar multiple of $[1, Y]$, then $y = xY$, so the local description of the chart is (x, Y) (which $U_X \cap \text{Bl}_0 \mathbb{A}^2 \cong \mathbb{A}^2$). And, under the projection $\varphi : U_X \cap \text{Bl}_0 \mathbb{A}^2 \rightarrow \mathbb{A}^2$, one has $\varphi(x, Y) = (x, xY) = (x, y) = \bar{x}$.

Which, the equation pulls back to $x^4 - x^3 + x^4 Y^4 + x^2 Y^2 = x^2(x^2 - x + x^2 Y^4 + Y^2) = 0$. With x^2 defines the exceptional divisor, the equation is $f_X = x^2 - x + x^2 Y^4 + Y^2 = 0$. For any singular points, it has the following relations:

$$\begin{aligned} \frac{\partial f_X}{\partial x} &= 2x - 1 + 2xY^4 = 0 \implies 2x(1 + Y^4) = 1 \\ \frac{\partial f_X}{\partial Y} &= 4x^2 Y^3 + 2Y = 0 \implies 2Y(2x^2 Y^2 + 1) = 0 \end{aligned} \tag{6.1}$$

This indicates that $x \neq 0$ and $Y^4 \neq -1$ (for the first equation to hold). There are two cases:

First, if $Y = 0$, then plugin the first partial reads $2x = 1$, or $x = \frac{1}{2}$. Yet, plugin $(x, Y) = (\frac{1}{2}, 0)$ to f_X , it reads $\frac{1}{2^2} - \frac{1}{2} + 0 = -\frac{1}{4} \neq 0$, so such point is not in the proper transform.

Second, if $2x^2 Y^2 + 1 = 0$ (or $2x^2 Y^2 = -1$), then both $x, Y \neq 0$, and one reads $Y^2 = -\frac{1}{2x^2}$, and one also has $1 + Y^4 = \frac{1}{2x}$, or $Y^4 = \frac{1}{2x} - 1$. Plugin to f_X , it simplifies to the following:

$$\begin{aligned}
f_X &= x^2 - x + x^2 Y^4 + Y^2 = x^2 - x + x^2 \left(\frac{1}{2x} - 1 \right) - \frac{1}{2x^2} \\
&= -\frac{1}{2x^2} (x^3 + 2) = 0
\end{aligned} \tag{6.2}$$

Hence, $x^3 + 2 = 0$, or $x^3 = -2$. Then, with $2x^2 Y^2 = -1$, $-4Y^2 = 2x^3 Y^2 = -x$, so $4Y^2 = x$. Plug in back to f_X , one yields $x^2 - x + x^2 Y^4 + Y^2 = x^2 - x + \frac{x^2}{16} + \frac{x}{4} = 0$, showing $\frac{x}{16}(17x - 12) = 0$. With $x \neq 0$, it must be $x = \frac{12}{17}$. Yet, this doesn't satisfy the equation $x^3 = -2$, which is a contradiction. Hence, in this case there can't have any singular points either.

Under the open subset U_Y (where $Y \neq 0$, again WLOG $Y = 1$), since (x, y) is some multiple of $[X, 1]$, then $x = yX$, so the local description of the chart (X, y) , and under the projection $\mu(X, y) = (yX, y)$.

Which, the equation pulls back to $y^4 X^4 - y^3 X^3 + y^4 + y^2 = y^2(y^2 X^4 - yX^3 + y^2 + 1) = 0$, with y^2 defines the exceptional divisor, the defining equation is $f_Y = y^2 X^4 - yX^3 + y^2 + 1 = 0$. For any singular points, it has the following relations:

$$\begin{aligned}
\frac{\partial f_Y}{\partial X} &= 4y^2 X^3 - 3yX^2 = 0 \implies yX^2(4yX - 3) = 0 \\
\frac{\partial f_Y}{\partial y} &= 2yX^4 - X^3 + 2y = 0
\end{aligned} \tag{6.3}$$

If $X = 0$ or $y = 0$ (to satisfy $\partial X = 0$), $X = 0$ implies $2y = 0$ (or $y = 0$), while $y = 0$ implies $-X^3 = 0$, $X = 0$. So, this directly implies $X = 0$ iff $y = 0$. Yet, plugin $(X, y) = (0, 0)$ for f_Y , one yields $1 \neq 0$, so these points are not on the proper transform, hence doesn't affect.

Else, if $4yX = 3$ (where $y, X \neq 0$), then the second partial tells that $2(2yX^4 - X^3 + 2y) = 3X^3 - 2X^3 + 4y = X^3 + 4y = 0$, so $4y = -X^3$. Plugin to the initial relation, $4yX = -X^4 = 3$, so $X^4 = -3$, and $X^3 = -4y$. Plugin to f_Y one yields $-3y^2 + 4y^2 + y^2 + 1 = 0$, or $2y^2 + 1 = 0$. However, this implies $y^2 = -\frac{1}{2}$, showing $X^6 = 16y^2 = 4$, so $X^{12} = 16$; yet $X^4 = -3$ implies $X^{12} = -27$, which reaches a contradiction. So, there cannot be any singular points in U_Y .

The above two claims showed that the Cusp in Exercise 5.1 (Problem 3) is non-singular after blowing-up.

Node: Its corresponding equation is $g = x^6 - xy + y^6 = 0$.

Under the open subset U_X , the chart has description (x, Y) , so map has description $\varphi(x, Y) = (x, xY)$. Then, the equation pulls back to $x^6 - x^2 Y + x^6 Y^6 = x^2(x^4 - Y + x^4 Y^6) = 0$. With x^2 defines the exceptional divisor, the equation is $g_X = x^4 - Y + x^4 Y^6 = 0$. Then, the singular points satisfy:

$$\frac{\partial g_X}{\partial x} = 4x^3(1 + Y^6) = 0, \quad \frac{\partial g_X}{\partial Y} = -1 + 6x^4 Y^5 = 0 \tag{6.4}$$

Hence, $6x^4 Y^5 = 1$, indicating $x, Y \neq 0$. So, one has $Y^6 = -1$ instead (for ∂x to hold). But, plugin g_X one yields $x^4 - Y - x^4 = -Y \neq 0$, showing this point is not in the proper transform, hence not a singular point.

Else, under the open subset U_Y , the chart has description (X, y) , with the map's description $\varphi(X, y) = (yX, y)$. Then, the equation pulls back to $y^6 X^6 - y^2 X + y^6 = y^2(y^4 X^6 - X + y^4) = 0$. With y^2 defines the exceptional divisor, the equation is $g_Y = y^4 X^6 - X + y^4 = 0$. Then, the singular points satisfy:

$$\frac{\partial g_Y}{\partial X} = 6y^4 X^5 - 1 = 0, \quad \frac{\partial g_Y}{\partial y} = 4y^3(X^6 + 1) = 0 \quad (6.5)$$

With $6y^4 X^5 = 1$, this indicates $X, y \neq 0$; so, for the second equation to hold, $X^6 = -1$. Yet, plugin to g_Y , one yields $-y^4 - X + y^4 = -X \neq 0$, which is not in the proper transform, hence not a singular point.

As a result, the two charts contained no singular points (for the proper transform), showing the blowing-up is non-singular.

- (b) If consider a plane curve with a node (WLOG, say up to a shift of coordinate, the node $P = (0, 0)$), if say that the equation is given by $f = f_2 + \dots + f_d$ (each f_i is a homogeneous polynomial of degree i), then the description of a node provides that $f_2 = ax^2 + bxy + cy^2 = (ex + fy)(gx + hy)$, where $(ex + fy)(gx + hy)$ are distinct linear factors. If view $k[x, y]$ as a k -vector space, the two being distinct linear factors imply they're linearly independent, hence one must have $eh - fg \neq 0$.

If consider the blowing-up as a proper transform in $\text{Bl}_0 \mathbb{A}^2$, for simplicity let's derive some statements using the open subset U_X of $\text{Bl}_0 \mathbb{A}^2$ (where $X \neq 0$). In here the local chart is (x, Y) , with map $\varphi(x, Y) = (x, xY)$ projecting down to the original affine space. Notice that for equation f , any of the homogeneous polynomial $f_i = \sum_{j=0}^i a_{j,i} x^j y^{i-j}$ pulls back to $\sum_{j=0}^i a_{j,i} x^j \cdot x^{i-j} Y^{i-j} = x^i \cdot \sum_{j=0}^i Y^{i-j}$. With $i \geq 2$ for f , define $\bar{f}_i := x^{i-2} \sum_{j=0}^i a_{j,i} Y^{i-j}$, then f pulls back to $x^2(\bar{f}_2 + \dots + \bar{f}_d)$ (since each f_i pulls back to $x^2 \cdot \bar{f}_i$).

Notice that $\varphi^{-1}(P)$ (under this affine chart) all have x -coordinate being 0 (since one needs $\varphi(x, Y) = (x, xY) = P = (0, 0)$). With all the $i \geq 3$ term including a factor of x in $\bar{f}_i$, then at any point in $\varphi^{-1}(P)$, $\bar{f}_i$ vanishes for $i \geq 2$, with only $\bar{f}_2 = (e + fY)(g + hY)$ left (this is constructed by the previous definition). Which, in case for $\bar{f} = 0$, we need $\bar{f}_2 = 0$. Which, there are two possible cases:

- If both $f, h \neq 0$, $\bar{f}_2$ is a quadratic, then the solutions yield $Y = -\frac{e}{f}, -\frac{g}{h}$. Which, under the blowing-up $\text{Bl}_0 \mathbb{A}^3$, they correspond to the matrix $\begin{pmatrix} x & y \\ X & Y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -\frac{e}{f} \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 1 & -\frac{g}{h} \end{pmatrix}$ (these two are distinct, based on the fact that $eh - fg \neq 0$, or $\frac{e}{f} \neq \frac{g}{h}$).
Notice that at these points $\frac{\partial \bar{f}}{\partial Y} \neq 0$, since $\frac{\partial \bar{f}_2}{\partial Y} = (eh + fg) - 2fhY$. Which, if $Y = -\frac{e}{f}$, it provides $-eh + fg = -(eh - fg) \neq 0$; else if $Y = -\frac{g}{h}$, it provides $eh - fg \neq 0$. So, at any point of $\varphi^{-1}(P)$, in this case the partial derivative of Y is non-vanishing, hence non-singular.
- If one of f, h is 0, then $\bar{f}_2$ is linear. If $f = 0$, for linear independence of $(ex + fy), (gx + hy)$, it enforces $h \neq 0$, then the only point is $Y = -\frac{g}{h}$; else if $h = 0$, by similar reason it enforces $f \neq 0$, then the only point is $Y = -\frac{e}{f}$. Which, regardless these two cases both yield non-singular points.

But, in both cases we still need to worry about the other affine chart (if they provide more information or not): Using the same machinery, if looking into the open subset U_Y in $\text{Bl}_0 \mathbb{A}^2$ (where $X \neq 0$), the local description is by (X, y) , and with the projection $\varphi(X, y) = (yX, y)$. So, the equation f pulls back to an equation $\bar{\bar{f}}$, where $y^2 \cdot \bar{\bar{f}} = y^2 \cdot (\bar{f}_2 + \dots + \bar{f}_d)$, where each $\bar{\bar{f}}_i = y^{i-2} \sum_{j=0}^i b_{j,i} X^j$ (here the coefficients depend on each f_i). So, with $\varphi^{-1}(P)$ having the y -coordinate being 0, then every term $\bar{\bar{f}}_i$ vanishes for $i \geq 3$ (since they contained nontrivial factor of y).

Then, looking at the factor $\bar{\bar{f}}_2 = (eX + f)(gX + h)$, one can consider the two cases:

- If $e, g \neq 0$, then one has $X = -\frac{f}{e}, X = -\frac{h}{g}$. Which, in $\text{Bl}_0 \mathbb{A}^2$ they correspond to the matrices $\begin{pmatrix} x & y \\ X & Y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -\frac{f}{e} & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ -\frac{h}{g} & 1 \end{pmatrix}$. (Note: in this case (∂x) never vanishes at these points, using the same logic as the proof in U_X).

- If $e = 0$ or $g = 0$, then first case corresponds to $X = -\frac{h}{g}$, the second case corresponds to $X = -\frac{f}{e}$ (since again by linear independence of factors $(ex + fy), (gx + hy)$, $e = 0$ implies $g \neq 0$, while $g = 0$ implies $e \neq 0$).

For the case where $f, h \neq 0$, in the chart U_X one has $\varphi^{-1}(P) = \left\{ \begin{pmatrix} 0 & 0 \\ 1 & -\frac{e}{f} \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & -\frac{g}{h} \end{pmatrix} \right\}$. If $e \neq 0$, then the point $X = -\frac{f}{e}$ in U_Y reads $\begin{pmatrix} 0 & 0 \\ -\frac{f}{e} & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -\frac{e}{f} \end{pmatrix}$ (since $[X, Y] \in \mathbb{P}^1$, so up to scaling they're the same). Else if $f \neq 0$, then the point $X = -\frac{h}{g}$ reads $\begin{pmatrix} 0 & 0 \\ -\frac{h}{g} & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -\frac{g}{h} \end{pmatrix}$. So, in this case U_Y doesn't provide extra points, it's indeed the two points listed in U_X .

For the case where $f = 0$ or $h = 0$ (for definiteness say $f = 0$), it implies $h \neq 0$, and one has $\varphi^{-1}(P) = \left\{ \begin{pmatrix} 0 & 0 \\ 1 & -\frac{g}{h} \end{pmatrix} \right\}$ in U_X . for $(ex + fy), (gx + hy)$ to be linearly independent, one needs $e \neq 0$, which extends to two cases: Either $g = 0$ or $g \neq 0$. If $g = 0$, then in U_Y the solution provides $X = -\frac{f}{e} = 0$, which corresponds to $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ (distinct from $\varphi^{-1}(P)$); else if $g \neq 0$, then in the solution it has $X = -\frac{f}{e} = 0$ and $X = -\frac{h}{g} \neq 0$, corresponding to the two matrices $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ -\frac{h}{g} & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -\frac{g}{h} \end{pmatrix}$. Which, we yield two distinct points, with one of them coincide with the solution derived in U_X .

So, regardless we'll always find two distinct solutions, and they're non-singular because under the affine chart, one of the partial derivative doesn't vanish.

- (c) Given the equation of the Tacnode being $x^4 - x^2 + y^4 = 0$ in Exercise 5.1 (Problem 3). Take the BLow-up $\tilde{Y}$ as the proper transform under $\text{Bl}_0 \mathbb{A}^2$, and take the open subset U_Y (where $Y \neq 0$), then the local description of the chart is (X, y) , with $\varphi(X, y) = (yX, y)$.

Then, the defining equation pulls back to $y^4 X^4 - y^2 X^2 + y^4 = y^2(y^2 X^4 - X^2 + y^2) = 0$, so with y^2 describes the exceptional divisor, the defining equation is $y^2 X^4 - X^2 + y^2 = 0$.

Notice that $\varphi^{-1}(P)$ yields $(X, y) = (0, 0)$. Which, since under the open subset U_Y the defining equation has the lowest degree nonzero homogeneous polynomial being $-X^2 + y^2 = (y - X)(y + X)$, showing that the point $\varphi^{-1}(P)$ (under this chart) has two distinct linear factors corresponding to f_2 (the lowest degree nonzero homogeneous polynomial of the defining equation), hence it's a node.

For safety (meaning $\varphi^{-1}(P)$ is only one point), let's check the other open subset U_X (where $X \neq 0$). The local description of the chart is (x, Y) , with the map $\varphi(x, Y) = (x, xY)$. Which, the defining equation pulls back to $x^4 - x^2 + x^4 Y^4 = x^2(x^2 - 1 + x^2 Y^4) = 0$, with x^2 defines the exceptional divisor, the equation is $x^2 - 1 + x^2 Y^4 = 0$. Notice that $\varphi^{-1}(P)$ should yield $(x, Y) = (0, 0)$, but this point is not on the curve defined by the equation mentioned, hence $\varphi^{-1}(P)$ doesn't exist in this chart. This shows the only point in $\varphi^{-1}(P)$ lies in U_Y (which has been calculated above), and it is a node.

- (d) First, the point $(0, 0)$ in $x^5 - y^3 = 0$ has multiplicity 3, since the lowest degree homogeneous polynomial is $-y^3$, indicating that it's a triple point.

If blowing up at $O = (0, 0)$ (viewed as the proper transform in $\text{Bl}_0 \mathbb{A}^2$), take the open subset U_Y (where $Y \neq 0$), the local chart provides (X, y) , and the map can be defined as $\varphi(X, y) = (yX, y)$. So, the defining equation pulls back to $y^5 X^4 - y^3 = y^3(y^2 X^4 - 1) = 0$. With y^3 defines the exceptional divisor, $y^2 X^4 - 1 = 0$ is the defining equation. Which, $\varphi^{-1}(O)$ supposedly provides $(X, y) = (0, 0)$, yet this point is not on the algebraic set (since plugin provides $-1 \neq 0$), so we need to see the other affine chart.

Take U_X (where $X \neq 0$), the local chart provides (x, Y) , and the map can be defined as $\varphi(x, Y) = (x, xY)$. So, the defining equation pulls back to $x^5 - x^3 Y^3 = x^3(x^2 - Y^3) = 0$. So, the defining equation is given by $x^2 - Y^3 = 0$. Here, $\varphi^{-1}(O)$ provides $(x, Y) = (0, 0)$, which is on the locus; and, notice that the lowest degree homogeneous polynomial is x^2 , showing that $\varphi^{-1}(O)$ is a double point (and since the linear factor of x^2 is not distinct, it's

not a node, must be a Tacnode or a Cusp). If limit to $k = \mathbb{R}$, it defines a cuspidal cubic, which the point $\varphi^{-1}(O)$ is a cusp.

Finally, under this affine chart (isomorphic to $\mathbb{A}^2$), the blowing-up of $x^2 - Y^3 = 0$ at $(x, Y) = (0, 0)$ as a set, can be viewed as the proper transform in $\text{Bl}_0 \mathbb{A}^2 = \{(\bar{x}, [\ell])\}$, where $\bar{x} = (x, Y)$ and $[\ell] = [X', Y']$ has $\begin{pmatrix} x & Y \\ X' & Y' \end{pmatrix}$ with rank ≤ 1 .

On the open subset $V_{X'}$ (where $X' \neq 0$), the local chart is (x, Y') , with map $\mu(x, Y') = (x, xY')$ as the projection back to the affine chart. Which, the equation pulls back to $x^2 - x^3 Y'^3 = x^2(1 - xY'^3) = 0$, which the x^2 is defining the exceptional divisor, so the equation under this set $V_{X'}$ is $1 - xY'^3 = 0$. But, notice that $\mu^{-1}(0, 0)$ supposedly provides $(x, Y') = (0, 0)$, which is not on the curve. Hence, we need to look at the other open subset.

Take $V_{Y'}$ (where $Y' \neq 0$), the local chart is (X', Y) , with projection $\mu(X', Y) = (YX', Y)$, so the equation pulls back to $Y^2 X'^2 - Y^3 = Y^2(X'^2 - Y) = 0$. With Y^2 defines the exceptional divisor, the equation is $X'^2 - Y = 0$. Notice that this has linear factor, hence under this chart the point $\mu^{-1}(0, 0)$ (being $(X', Y) = (0, 0)$) is non-singular, showing the second blowing up in fact resolves the singularity at the point.

7 D

Problem 7

Hartshorne 5.7:

Let $Y \subseteq \mathbb{P}^2$ be a non-singular plane curve of degree > 1 , defined by the equation $f(x, y, z) = 0$. Let $X \subseteq \mathbb{A}^3$ be the affine variety defined by f (this is the cone over Y ; see (Exercise 2.10)). Let P be the point $(0, 0, 0)$, which is the *vertex* of the cone. Let $\varphi: \tilde{X} \rightarrow X$ be the blowing-up of X at P .

- (a) Show that X has just one singular point, namely P .
- (b) Show that $\tilde{X}$ is non-singular (cover it with open affines).
- (c) Show that $\varphi^{-1}(P)$ is isomorphic to Y .

Solution:

- (a) Since Y is a projective algebraic set, its defining equation f is homogeneous of some degree d ; also, it is assumed to be non-singular, then under the affine cover of $\mathbb{P}^2$, the defining equation of Y in each of the affine chart is nonzero:

For instance, in the affine chart U_x (where $x \neq 0$), the defining equation of $Y \cap U_x \subseteq U_x$ is $f(1, y, z)$ (since can assume $x = 1$). Then, Y is non-singular implies $\frac{\partial f}{\partial y}(1, y, z)$ or $\frac{\partial f}{\partial z}(1, y, z)$ is nonzero (for definiteness, say $\frac{\partial f}{\partial y}(1, y, z) \neq 0$). Hence, in $\mathbb{A}^3$, if $(x, y, z) \in X$ has $x = a \neq 0$, then since $\frac{\partial f}{\partial y}$ has degree $d - 1$ (Note: in **Problem 5** it's proven that any homogeneous polynomial of degree d has nonzero partial derivatives being homogeneous of degree $d - 1$), then one has $\frac{\partial f}{\partial y}(a, y, z) = a^{d-1} \frac{\partial f}{\partial y}(1, \frac{y}{a}, \frac{z}{a}) \neq 0$. So, the partial derivative is nonvanishing at $(x, y, z) = (a, y, z) \neq 0$.

Apply similar logic to $y \neq 0$ and $z \neq 0$ (which instead need the affine chart $U_y \subseteq \mathbb{P}^2$ and $U_z \subseteq \mathbb{P}^2$ respectively, and the same claim on the partial derivatives in the affine chart), one can deduce that if $x \neq 0$, $y \neq 0$, or $z \neq 0$, at least one of the partial derivatives of f is nonvanishing. Hence, the only possible singularity must be at $P = (0, 0, 0)$.

Finally, the reason why P is singular, is because $Y \subset \mathbb{P}^2$ is of degree $d > 1$ (the homogeneous degree of f), hence all of its partial derivatives $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$ are homogeneous of degree $d - 1 \geq 1$. Hence, plugin $(x, y, z) = (0, 0, 0)$, since each monomial (of the partial derivatives) is non-constant, it must evaluate to be 0, showing all partial derivatives vanish at $P = (0, 0, 0)$.

- (b) We'll consider $\tilde{X}$ as the proper transform of X in $\text{Bl}_0 \mathbb{A}^3$. For simplicity, we'll just demonstrate one affine chart calculation: Pick $U_{X'}$ (where $X' \neq 0$ in $\text{Bl}_0 \mathbb{A}^3$, where the points are of the form $(\bar{x}, [\ell])$, where $\bar{x} = (x, y, z)$ and $[\ell] = [X', Y, Z]$ have $\begin{pmatrix} x & y & z \\ X' & Y & Z \end{pmatrix}$ with rank ≤ 1). Then, the local description is (x, Y, Z) .

Take the pull back of the equation, we in fact find $f(x, xY, xZ) = x^d \cdot f(1, Y, Z)$ (this is based on the fact that f is homogeneous of degree d). Here, since the preimage of $P = (0, 0, 0)$ in this affine chart has $x = 0$, $x^d = 0$ is the defining equation for the exceptional divisor, hence we're left with the equation $f(1, Y, Z)$ that defines $\tilde{X}$ in $U_{X'}$. Then, the partial derivatives with respect to Y, Z are given by $\frac{\partial f}{\partial Y}(1, Y, Z)$, and $\frac{\partial f}{\partial Z}(1, Y, Z)$. In part (a) we've verified that at least one of the two must be nonzero, showing $\tilde{X}$ is non-singular within the affine chart.

Also, notice that the same idea applies to all the other affine charts (like U_Y where $Y \neq 0$, and U_Z where $Z \neq 0$), showing that under the affine cover $\tilde{X}$ contains only non-singular points, hence non-singular.

- (c) It suffices to check that under each of the three standard affine charts of $\text{Bl}_0 \mathbb{A}^3$ and $\mathbb{P}^2$ (while viewing $\tilde{X}$ as the proper transform of X), it induces an isomorphism between $\varphi^{-1}(P)$ and Y .

Let $U_{X'} \cong \mathbb{A}^3$ denotes the affine open chart of $\text{Bl}_0 \mathbb{A}^3$ with $X' \neq 0$ (which has local description (x, Y, Z) and projection back to $\mathbb{A}^3$, by $\varphi(x, Y, Z) = (x, xY, xZ)$), and let $V_x \cong \mathbb{A}^2$ denotes the affine open chart of $\mathbb{P}^2$ where $x \neq 0$ (use $[x, y, z]$ as the coordinates of $\mathbb{P}^2$).

Which, the answer in part (a) suggested that in V_x (with coordinate (y, z) , since WLOG can assume $x = 1$), Y is cut out by the equation $f(1, y, z) = 0$; on the other hand the answer in part (b) suggested that in $U_{X'}$ (with coordinate (x, Y, Z)), the exceptional divisor is cut out by the equation $x^d = 0$, hence $\varphi^{-1}(P) \subseteq \{x = 0\} \subseteq U_{X'}$ (since if $\varphi(x, Y, Z) = (x, xY, xZ) = P = (0, 0, 0)$, one must have $x = 0$), which narrows down the the subset $\{(0, Y, Z) \in U_{X'}\} \cong \mathbb{A}^2$.

Also, notice that $\varphi^{-1}(P)$ under $U_{X'}$, is precisely the intersection $\{x^d = 0\} \cap (\tilde{X} \cap U_{X'})$ (since any point that projects down to $P = (0, 0, 0)$ must precisely have $x = 0$, which is precisely $\varphi^{-1}(P) \cap U_{X'}$ should contain this set; the other inclusion is by the fact that $\varphi^{-1}(P) \subseteq \tilde{X}$, and all point must have $x = 0$ when intersecting $U_{X'}$), while $\tilde{X} \cap U_{X'}$ has the defining equation $f(1, Y, Z)$ based on the result in part (b).

Hence, this shows the following isomorphism:

$$\begin{aligned} \varphi^{-1}(P) \cap U_{X'} &= \{x^d = 0\} \cap (\tilde{X} \cap U_{X'}) = \{(0, Y, Z) \mid f(1, Y, Z) = 0\} \\ &\cong \{(y, z) \mid f(1, y, z) = 0\} = Y \cap V_x \end{aligned} \tag{7.1}$$

Which, under the intersection with affine charts, one can see $\varphi^{-1}(P)$ is isomorphic to Y .

8 D

Problem 8

Hartshorne 5.11:

The Elliptic Quarter Curve in $\mathbb{P}^3$: Let Y be the algebraic set in $\mathbb{P}^3$ defined by the equations $x^2 - xz - yw = 0$ and $yz - xw - zw = 0$. Let P be the point $[x, y, z, w] = [0, 0, 0, 1]$, and let φ denote the projection from P to the plane $w = 0$. Show that φ induces an isomorphism of $Y \setminus \{P\}$ with the plane cubic curve $y^2z - x^3 + xz^2 = 0$ minus the point $[1, 0, -1]$. Then show that Y is an irreducible non-singular curve. It is called the *elliptic quartic curve in $\mathbb{P}^3$* . Since it is defined by two equations it is another example of complete intersection (Exercise 2.17).

Solution:

Let $Z := \{y^2z - x^3 + xz^2 = 0\} \subseteq \mathbb{P}^2$ be the algebraic set. Also, the projection $\varphi : \mathbb{P}^3 \setminus \{P\} \rightarrow \mathbb{P}^2$ is defined by $\varphi([x, y, z, w]) = [x, y, z]$ (here it rules out $P = [0, 0, 0, 1]$, so the projection is well-defined).

I. Projection $\varphi : Y \setminus \{P\} \rightarrow Z \setminus \{[1, 0, -1]\}$ is well-defined:

If restrict to $Y \setminus \{P\}$, we'll first show that $\varphi(Y \setminus \{P\}) \subseteq Z$: Given any $[x, y, z, w] \in Y$ that's not P , they satisfy the equations $\begin{cases} x^2 - xz - yw = 0 \\ yz - xw - zw = 0 \end{cases}$. Which, there are two cases:

- First, if $w = 0$, the equations reduce to $\begin{cases} x^2 - xz = 0 \\ yz = 0 \end{cases}$. So, one has $[x, y, z] = \varphi([x, y, z, w])$ satisfy:

$$y^2z - x^3 + xz^2 = y(yz) + (x^2 - xz)(z + x) = 0 \quad (8.1)$$

Hence, $\varphi([x, y, z, w]) = [x, y, z] \in Z$.

- Else, if $w \neq 0$, WLOG say $w = 1$. Which, the equations reduce to $\begin{cases} x^2 - xz - y = 0 \\ yz - x - z = 0 \end{cases}$, or $\begin{cases} x^2 = xz + y \\ yz = x + z \end{cases}$ (also, by rearrangement $xz = x^2 - y$ also). Again, one has $[x, y, z] = \varphi([x, y, z, w])$ satisfy:

$$\begin{aligned} y^2z - x^3 + xz^2 &= y(x + z) - x(xz + y) + z(x^2 - y) \\ &= xy + yz - x^2z - xy + x^2z - yz = 0 \end{aligned} \quad (8.2)$$

Hence again, $\varphi([x, y, z, w]) = [x, y, z] \in Z$.

These showed that as projection, $\varphi : Y \setminus \{P\} \rightarrow Z$ is well-defined.

Also, notice that for all $w \in k$, the point $[1, 0, -1, w] \notin Y$ (since plugin the polynomial $x^2 - xz - yw$, one has $1^2 - 1(-1) - 0 \cdot w = 2 \neq 0$ if $\text{char}(k) \neq 2$; if $\text{char}(k) = 2$, plugin the polynomial $yz - xw - zw$, one has $0 - 1 \cdot w - (-1) \cdot w = 0$). So $[1, 0, -1, w] \notin Y$, showing $[1, 0, -1] \notin \text{im}(\varphi)$ when restricting to $Y \setminus \{P\}$. So, $\varphi : Y \setminus \{P\} \rightarrow Z \setminus \{[1, 0, -1]\}$ is a well-defined projection.

II. Projection $\varphi : Y \setminus \{P\} \rightarrow Z \setminus \{[1, 0, -1]\}$ is an isomorphism:

For this, it suffices to find the inverse in some open neighborhood, for every point in $Z \setminus \{[1, 0, -1]\}$. Let U_y denotes the open set such that $y \neq 0$, and V denotes the open set where $x + z \neq 0$.

Notice that $Z \setminus \{[1, 0, -1]\} \subseteq U_y \cup V$, since if any $[x, y, z] \notin U_y \cup V$, one has $y = 0$ and $x + z = 0$ (or $x = -z$), which to have $[x, y, z] \in \mathbb{P}^2$ be well-defined, one needs $x = -z \neq 0$, hence WLOG assume $x = 1$, showing $[x, y, z] = [1, 0, -1]$, which is not contained in $Z \setminus \{[1, 0, -1]\}$. In other words, $Z \setminus \{[1, 0, -1]\}$ is fully contained in $U_y \cup V$, so having the inverse defined on these two open sets is enough.

Now, define the map $\psi : Z \setminus \{[1, 0, -1]\} \rightarrow Y \setminus \{P\}$ as follow:

$$\psi([x, y, z]) = \begin{cases} \left[x, y, z, \frac{x(x-z)}{y} \right], & y \neq 0 \\ \left[x, y, z, \frac{yz}{x+z} \right], & x + z \neq 0 \end{cases} \quad (8.3)$$

Which, the first condition can be defined on U_y , the second condition can be defined on V .

First, to show it's well-defined at each point, given any $[x, y, z] \in Z \setminus \{[1, 0, -1]\}$ that's also in $U_y \cap V$, notice that it satisfies the equation $y^2z - x^3 + xz^2 = 0$, or $y(yz) = x(x - z)(x + z)$ (where $y, x + z \neq 0$). So, one has $\frac{yz}{x+z} = \frac{x(x-z)}{y}$, showing the two definitions above agree, which the map ψ (into $\mathbb{P}^3$) is well-defined (and is a well-defined rational map on $Z \setminus \{[1, 0, -1]\}$).

Second, to show the image is in $Y \setminus \{P\}$, consider the following two cases:

- If $[x, y, z] \in (Z \setminus \{[1, 0, -1]\}) \cap U_y$, then $\psi([x, y, z]) = [x, y, z, \frac{x(x-z)}{y}]$ satisfies:

$$x^2 - xz - yw = x^2 - xz - y \cdot \frac{x(x-z)}{y} = x^2 - xz - x^2 + xz = 0 \quad (8.4)$$

$$yz - xw - zw = yz - (x+z) \cdot \frac{x(x-z)}{y} = \frac{1}{y}(y^2z - x^3 - xz^2) = 0 \quad (8.5)$$

(Note: the second equation is due to the fact that $[x, y, z] \in Z$, so $y^2z - x^3 - xz^2 = 0$).

Hence, $\psi([x, y, z]) \in Y$ based on the two equations.

- If $[x, y, z] \in (Z \setminus \{[1, 0, -1]\}) \cap V$ (in fact, V automatically rules out the point $[1, 0, -1]$), then $\psi([x, y, z]) = [x, y, z, \frac{yz}{x+z}]$ satisfies:

$$x^2 - xz - yw = (x^2 - xz) - y \cdot \frac{yz}{x+z} = \frac{1}{x+z}(x^3 - xz^2 - y^2z) = 0 \quad (8.6)$$

$$yz - xw - zw = yz - (x+z) \cdot \frac{yz}{x+z} = 0 \quad (8.7)$$

(Note: the first equation is due to $[x, y, z] \in Z$, so $y^2z - x^3 - xz^2 = 0$).

Hence, $\psi([x, y, z]) \in Y$ based on the two equations.

Also, since one of x, y, z is nonzero (since $[x, y, z] \in \mathbb{P}^2$), it's clear that $\psi([x, y, z]) \neq P = [0, 0, 0, 1]$, hence it's in $Y \setminus \{P\}$. So, $\psi : Z \setminus \{[1, 0, -1]\} \rightarrow Y \setminus \{P\}$ is well-defined.

Finally, to check it's the inverse of $\varphi : Y \setminus \{P\} \rightarrow Z \setminus \{[1, 0, -1]\}$, any $[x, y, z, w] \in Y \setminus \{P\}$ divides to the following two cases:

- If $y \neq 0$, then the first equation of Y , $x^2 - xz - yw = 0$ implies $w = \frac{x(x-z)}{y}$. Hence:

$$\psi \circ \varphi([x, y, z, w]) = \psi([x, y, z]) = \left[x, y, z, \frac{x(x-z)}{y} \right] = [x, y, z, w] \quad (8.8)$$

- Else if $x + z \neq 0$, then the second equation of Y , $yz - xw - zw = 0$ implies $w = \frac{yz}{x+z}$. Hence:

$$\psi \circ \varphi([x, y, z, w]) = \psi([x, y, z]) = \left[x, y, z, \frac{yz}{x+z} \right] = [x, y, z, w] \quad (8.9)$$

This shows that $\psi \circ \varphi = \text{id}$ on $Y \setminus \{P\}$. Also, it's clear that for all $[x, y, z] \in Z \setminus \{[1, 0, -1]\}$, one has $\varphi \circ \psi([x, y, z]) = \varphi([x, y, z, *]) = [x, y, z]$, showing $\varphi \circ \psi = \text{id}$ on $Z \setminus \{[1, 0, -1]\}$. Hence, the two sets are isomorphic.

III. Y is an irreducible Non-singular curve:

Based on the isomorphism above, it suffices to show that Z is a projective variety (which implies $Z \setminus \{[1, 0, -1]\}$ is an irreducible open subset of Z , so $Y \setminus \{P\}$ is irreducible, hence $Y = Y \setminus \{P\}$ under its subspace topology is also irreducible).

Which, to show Z is a projective variety, it's the most efficient to show its defining equation $y^2z - x^3 + xz^2$ is irreducible (hence the homogeneous ideal $(y^2z - x^3 + xz^2) \subset k[x, y, z]$ is prime, where Z as the algebraic set of this ideal, is a variety).

Since the polynomial has y -degree being 2, if it's reducible it must be the form $z \cdot y^2 + (xz^2 + x^3) = (f \cdot y - g)(h \cdot y - l)$, where $f, g, h, l \in k[x, z]$. Which, one has $f \cdot h = z$, showing both f, h

involves no x term, while one has z -degree 1. Hence up to unit, one can assume $f = z$ and $h = 1$ (so, $z \cdot y^2 + (xz^2 - x^3) = (z \cdot y - g)(y - l)$).

Then, in case to have no degree 1 y term, one must have $-z \cdot l - g = 0$, or $g = -z \cdot l$. Which, this implies $-zl^2 = gl = xz^2 - x^3$. Yet, this is clearly impossible, since plugin $z = 0, x = 1$, one yields $-zl^2 = 0$, while $xz^2 - x^3 = -1 \neq 0$. So, we reach a contradiction. Therefore, $y^2z - x^3 + xz^2$ cannot be reducible, hence irreducible, and this finishes the proof that Y is irreducible.

Finally, to show Y is non-singular, let's look at one of the affine open subset: Consider the open subset U_z (where $z \neq 0$), WLOG say $z = 1$. Hence, $Y \cap U_z$ can be defined by the two equations $\begin{cases} x^2 - x - yw \\ y^2 - x^3 - x \end{cases}$. So, the Jacobian matrix becomes (in terms of x, y, w from left to right columns, and the polynomials are from top to bottom row in order):

$$J = \begin{pmatrix} (2x - 1) & -w & -y \\ -(3x^2 - 1) & 2y & 0 \end{pmatrix} \quad (8.10)$$

Which, since $Y \subset \mathbb{P}^3$ has dimension at most 1 (since the two defining equations generate two hypersurfaces not containing each other, so the intersection must necessarily be a curve with one dimension lower), hence the Jacobian must have rank being the codimension $3 - 1 = 2$ for it to be non-singular. However, take $x = \frac{1}{2}, y = w = 0$, then $J = \begin{pmatrix} 0 & 0 & 0 \\ \frac{1}{4} & 0 & 0 \end{pmatrix}$, which clearly doesn't have rank 2. This shows that Y contains non-singular points, which finishes the last piece of the proof.